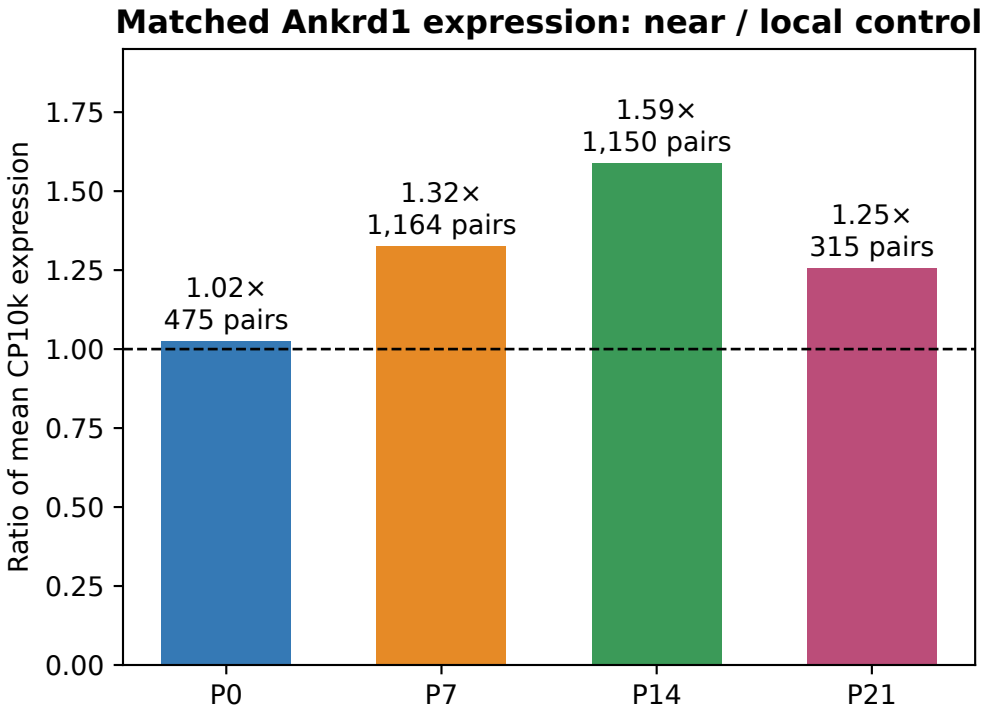
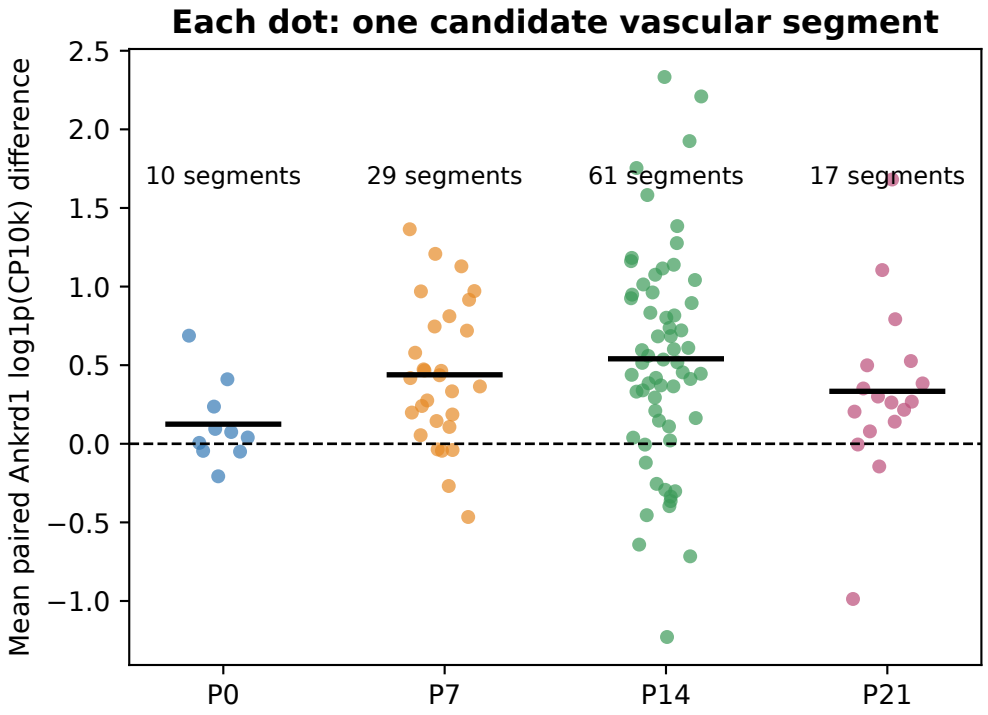
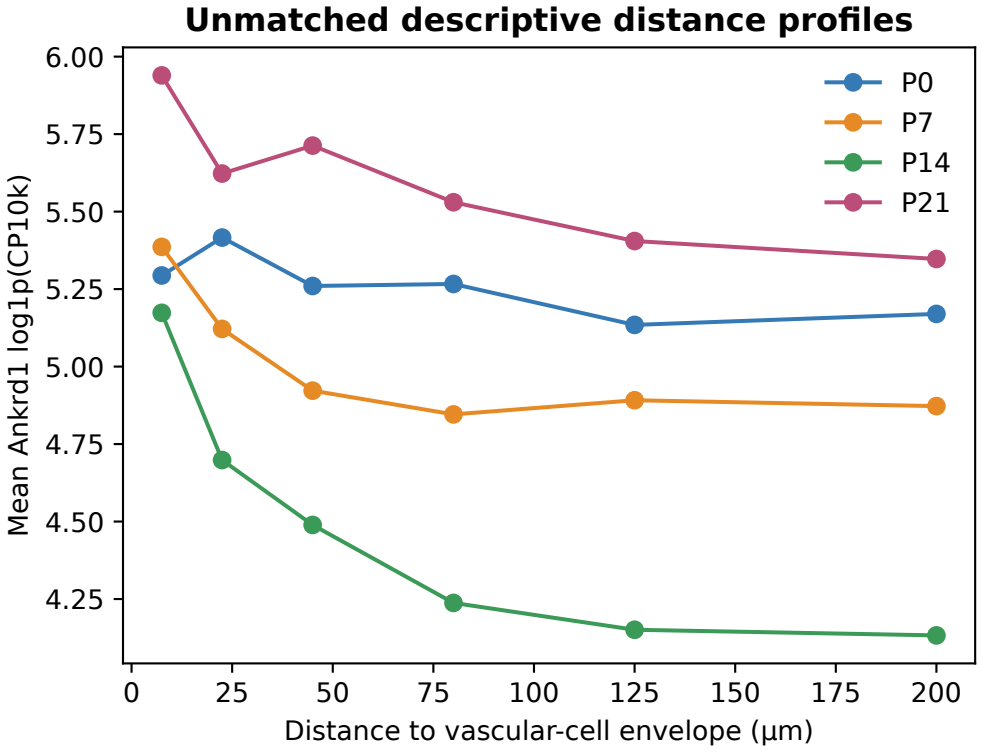
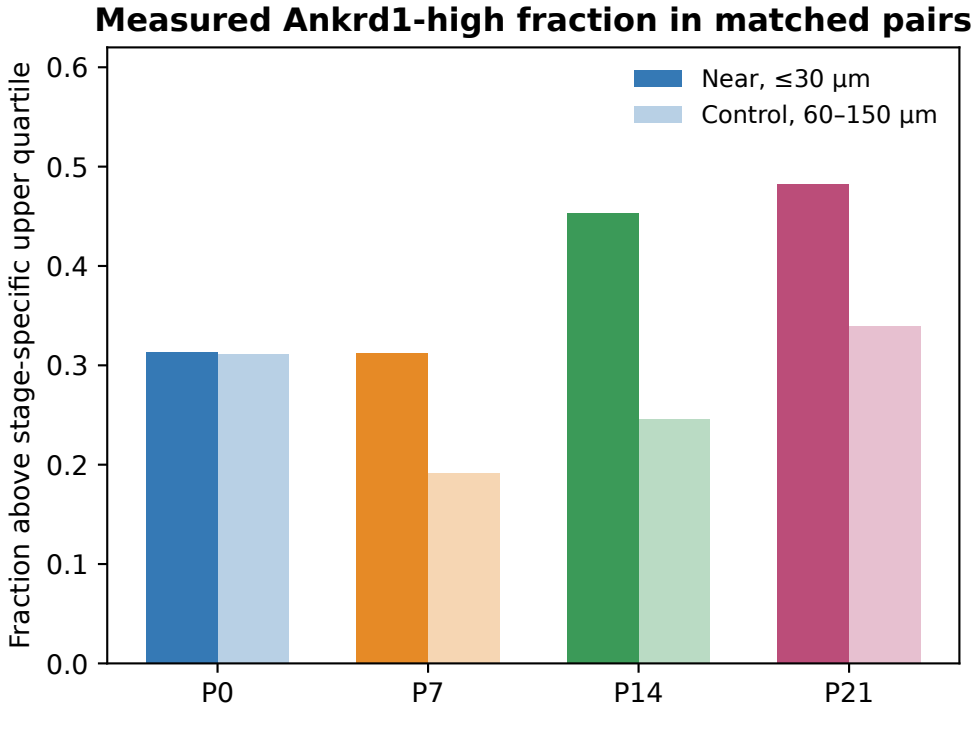
Ankrd1 and candidate arterial neighborhoods: P0–P21



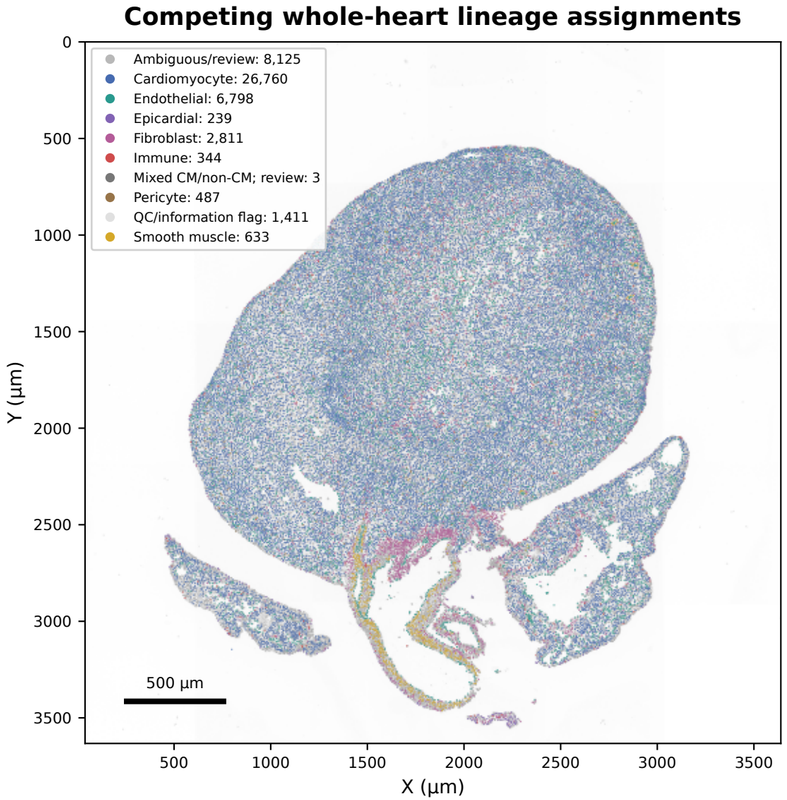


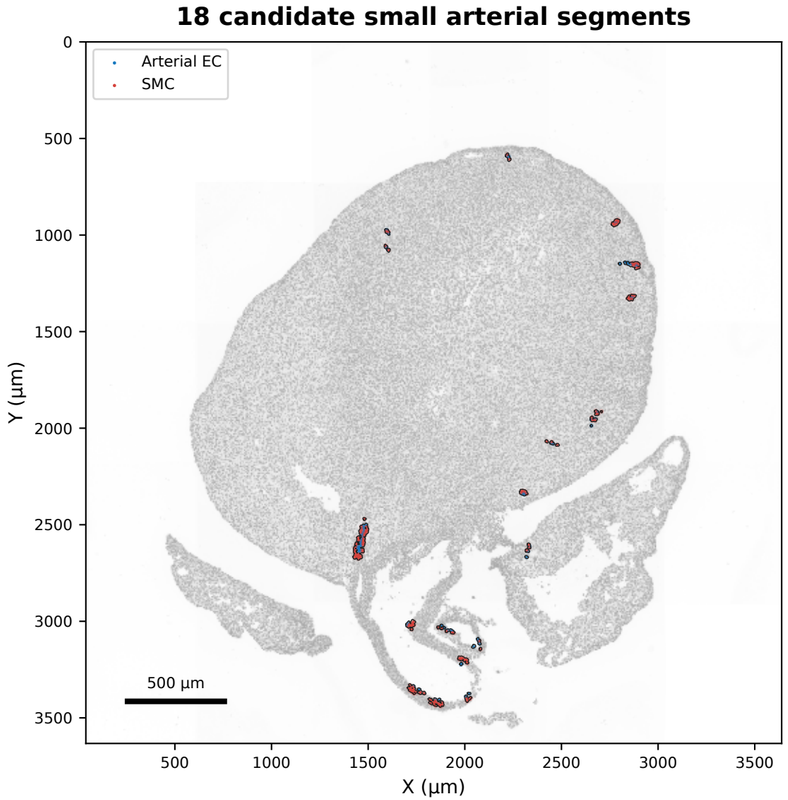


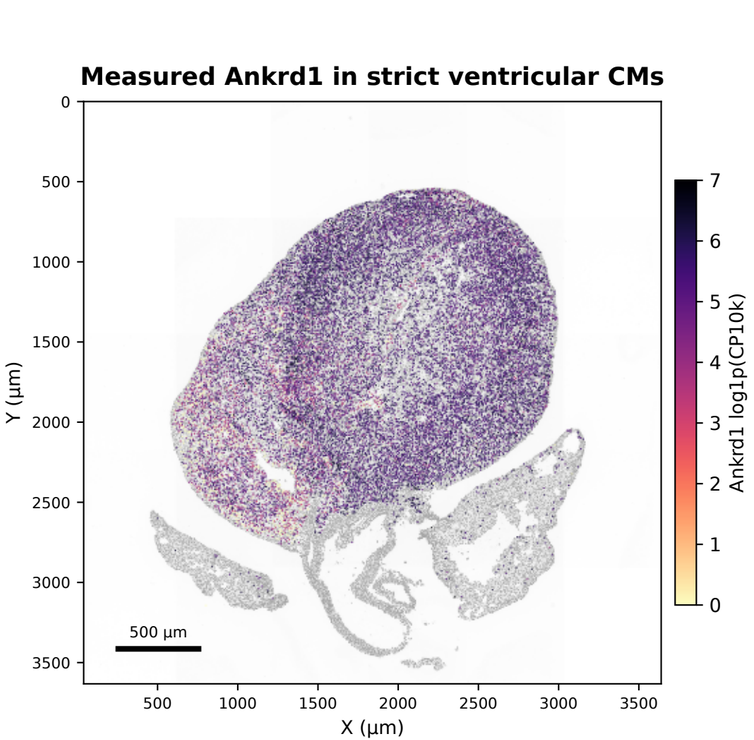


Primary: strict whole-reference ventricular CMs; independently defined arterial EC + SMC segments.
Near: 0–30 µm; local controls: 60–150 µm; 1:1 covariate matching within the same segment.
One section per age. Segment dots are exploratory within-section summaries, not independent animals.

P0: whole-heart reassessment and vascular context

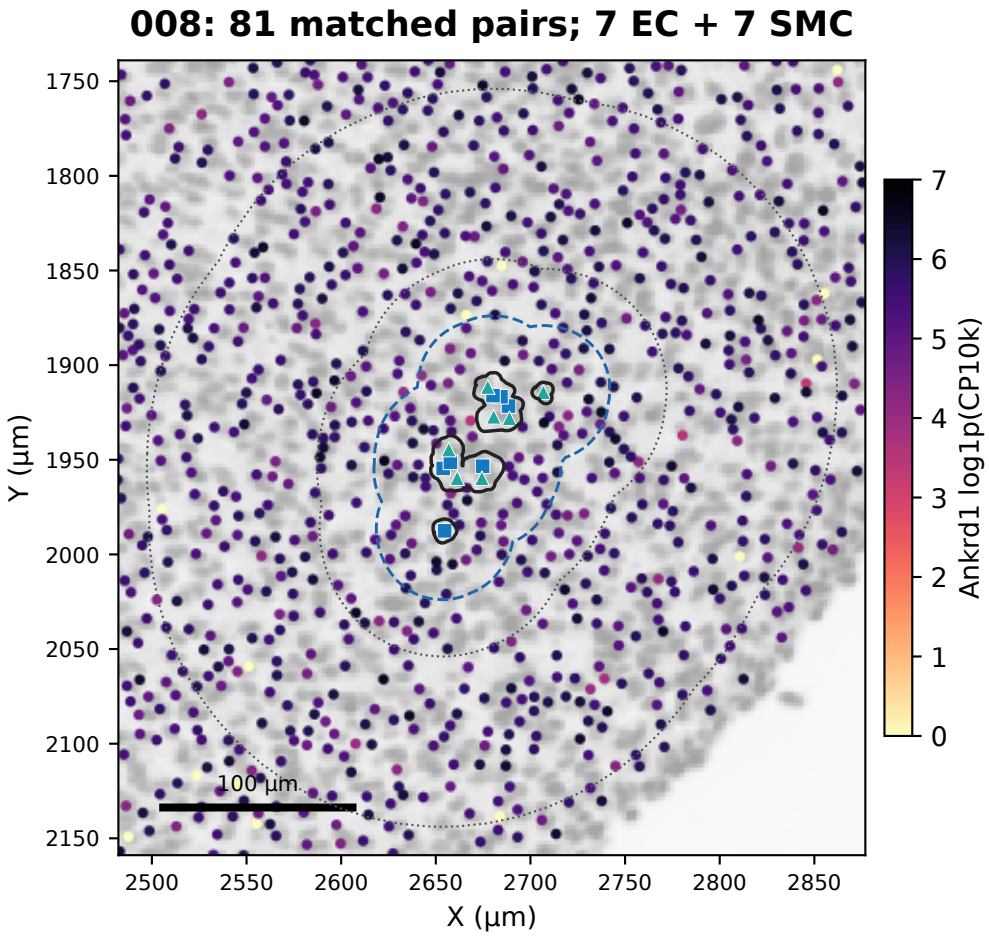


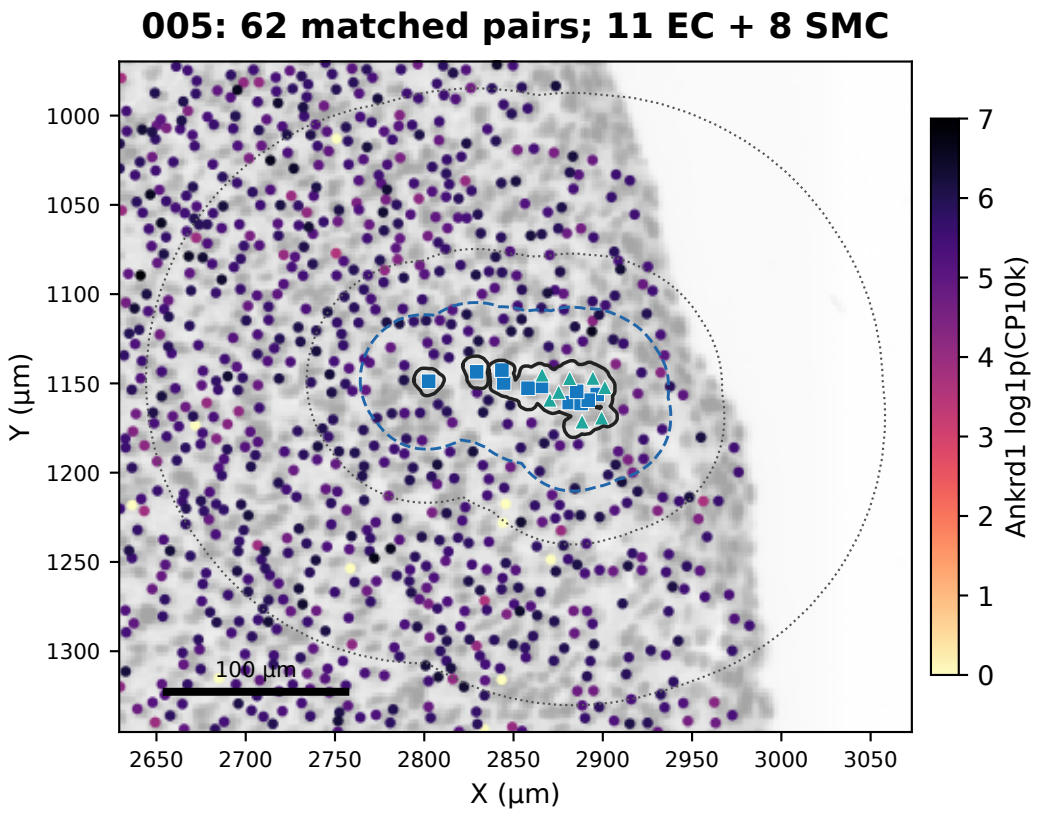


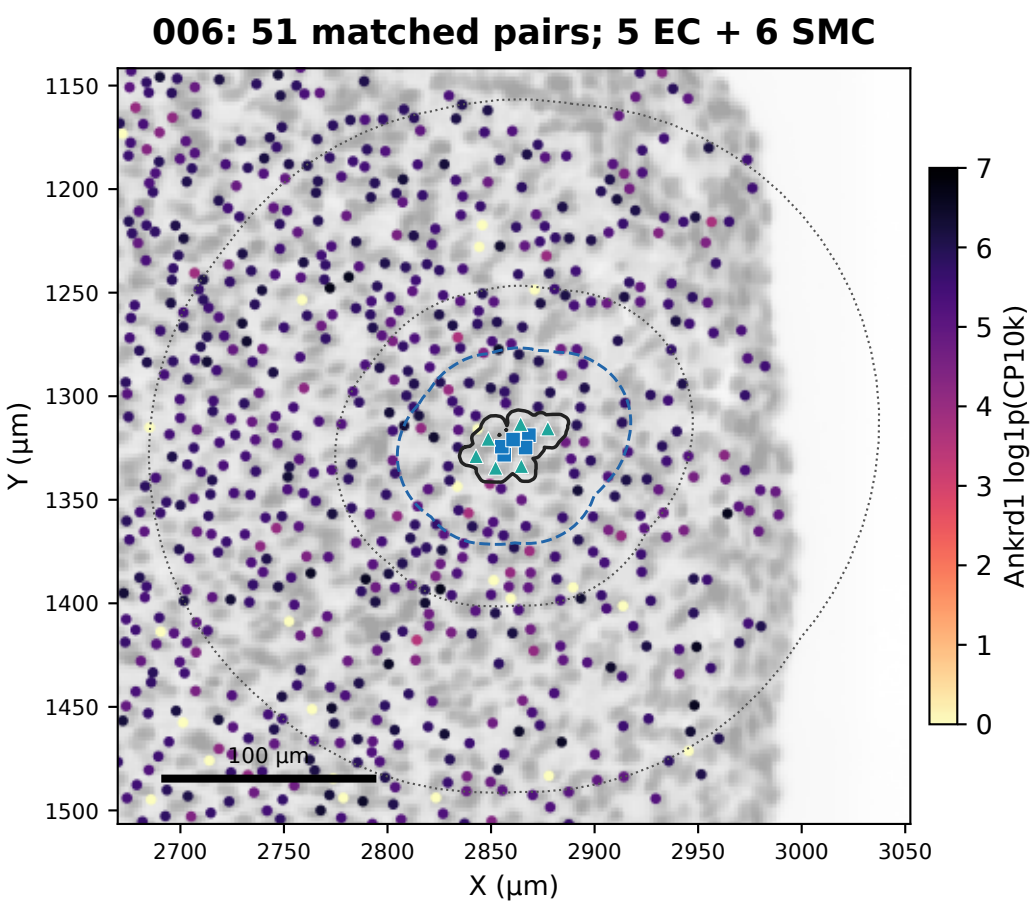


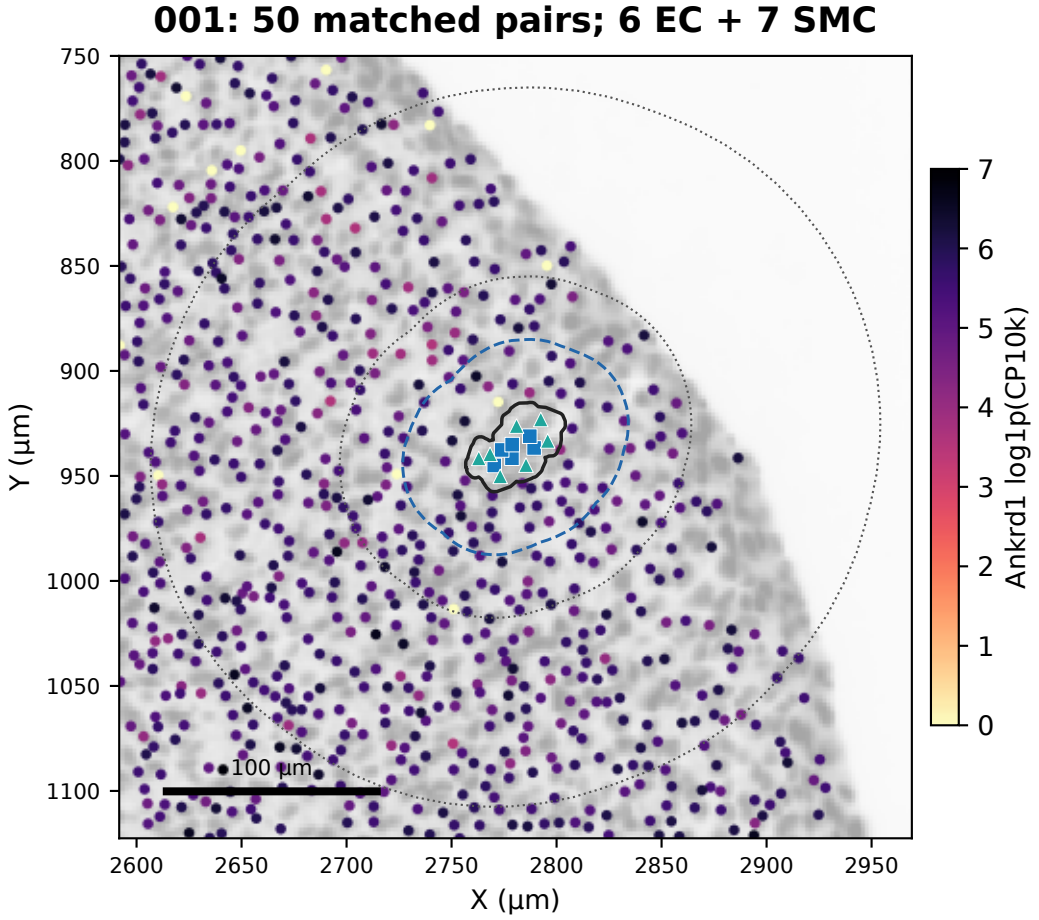
Original cell centroids and DAPI coordinates. All original cells retained; uncertain identities remain flagged.
Arterial candidates require arterial EC + surrounding smooth muscle; outlines are vascular-cell envelopes, not traced vessel walls.
Ankrd1, Creb5, Tlr4 and Atf3 were excluded from all new identity fits and fitting size-factor denominators.

P0: candidate vessel geometry and nearby ventricular CMs





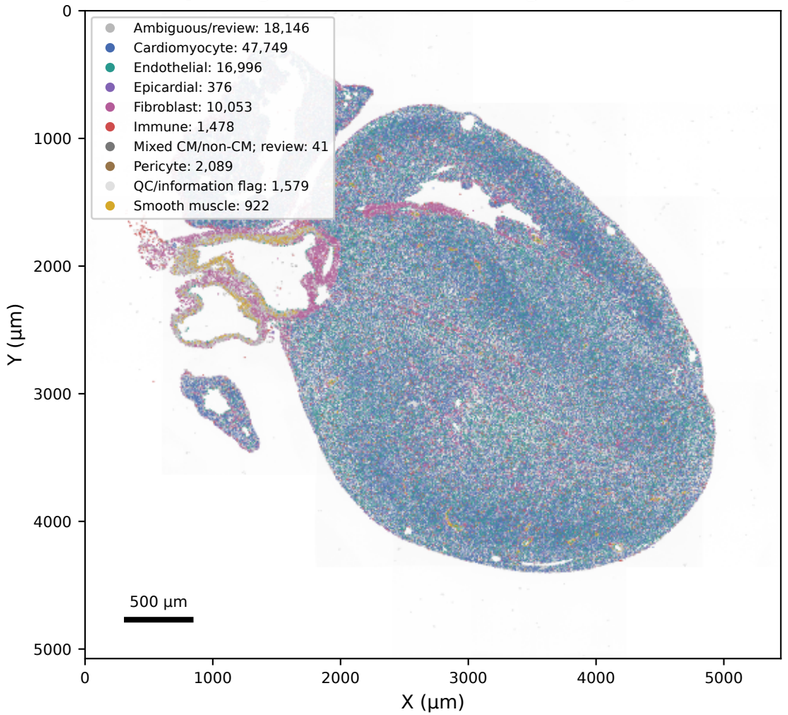




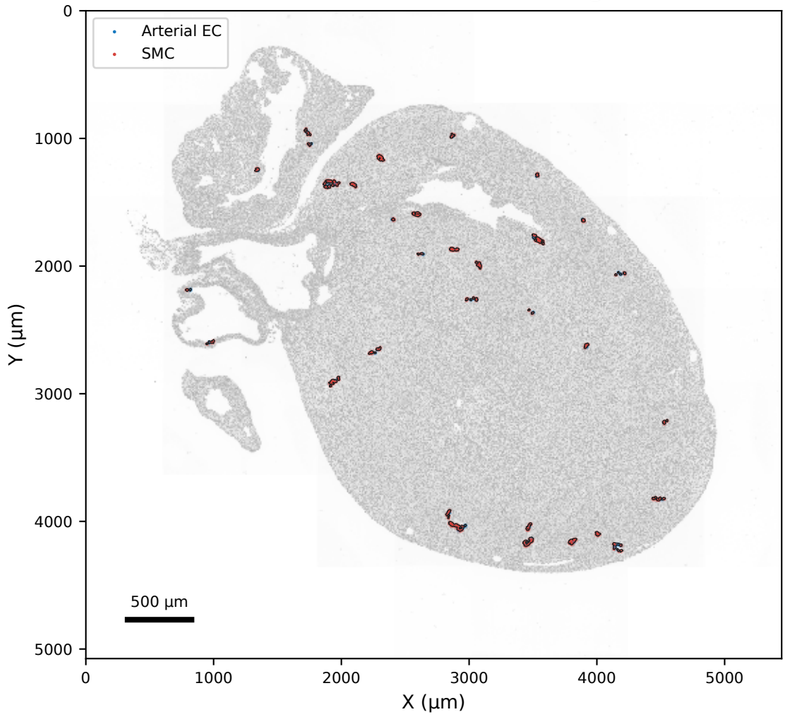
Examples selected by most matched pairs, without selecting on Ankrd1 effects. Square: arterial EC; triangle: SMC.
Black: vascular-cell envelope; blue dashed: 30 µm; gray dotted: 60 and 150 µm.
CM color shows directly measured expression, without smoothing or imputation. Geometry requires anatomical review.

P7: whole-heart reassessment and vascular context

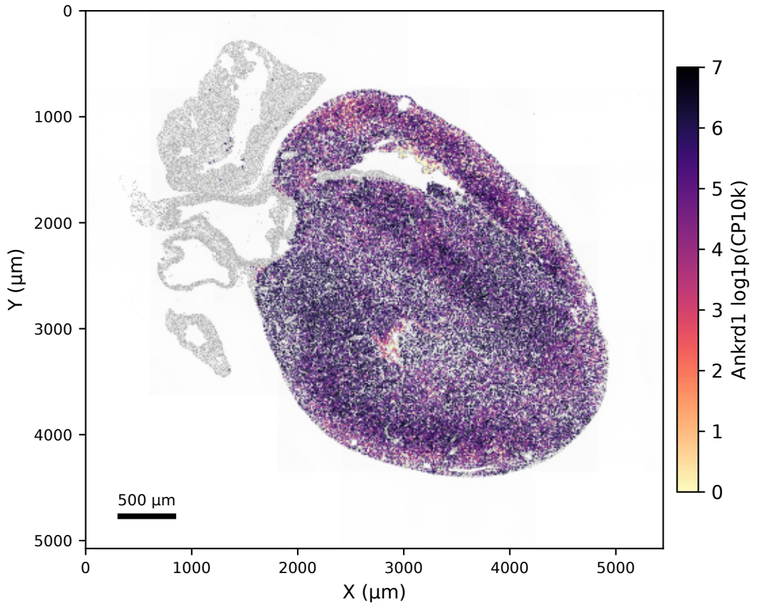
Competing whole-heart lineage assignments



33 candidate small arterial segments

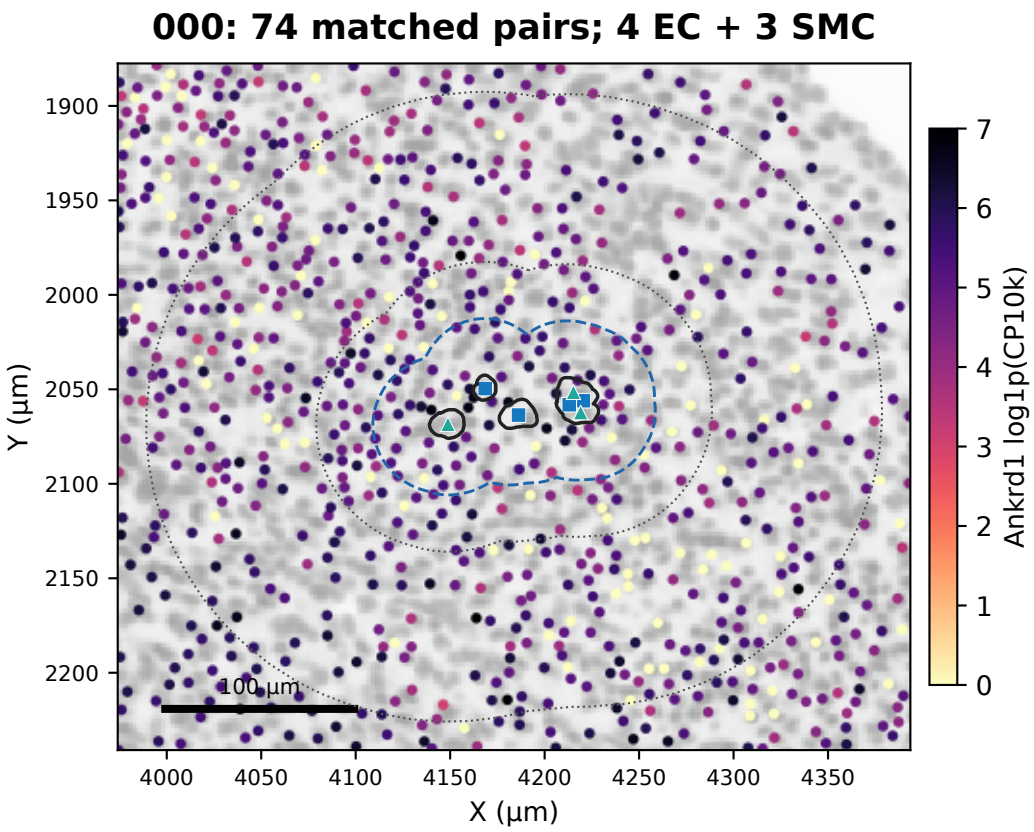


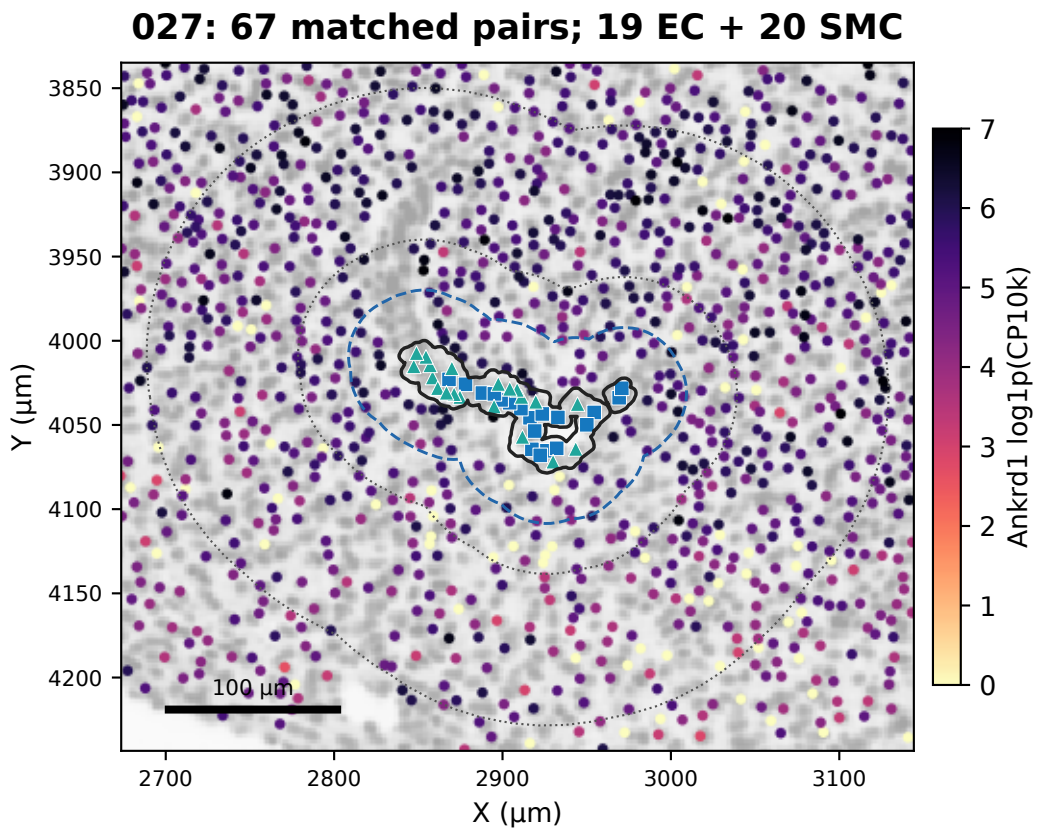
Measured Ankrd1 in strict ventricular CMs

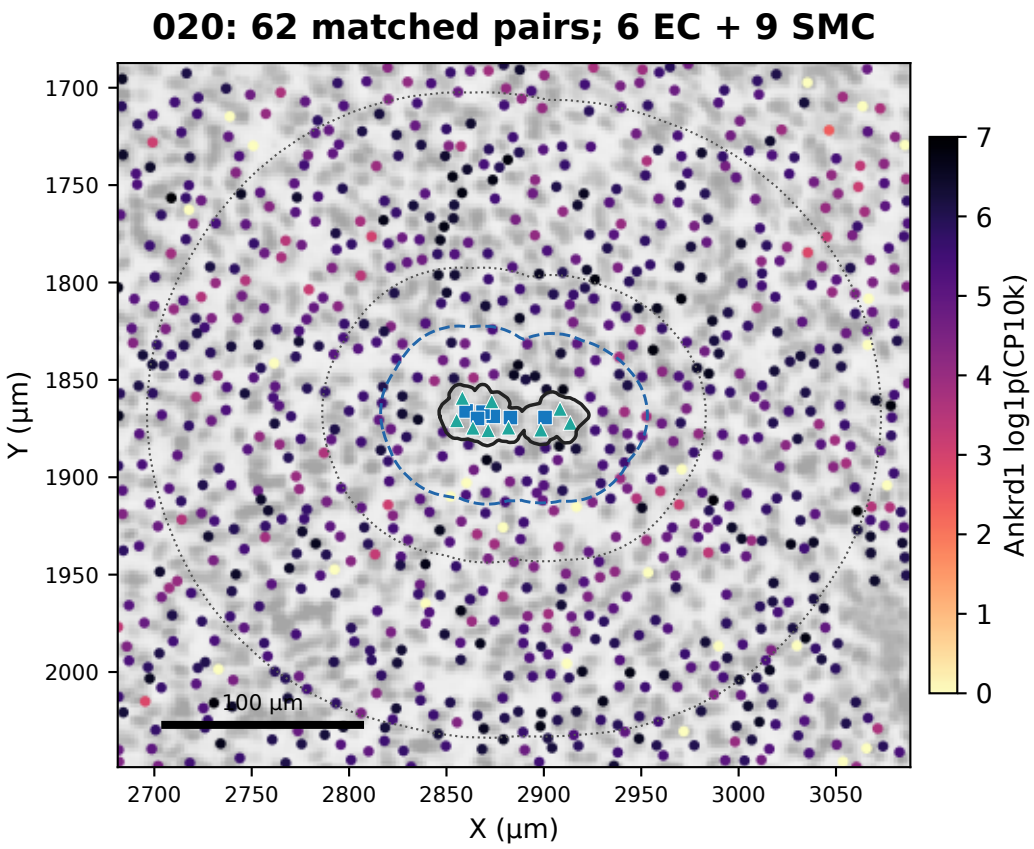


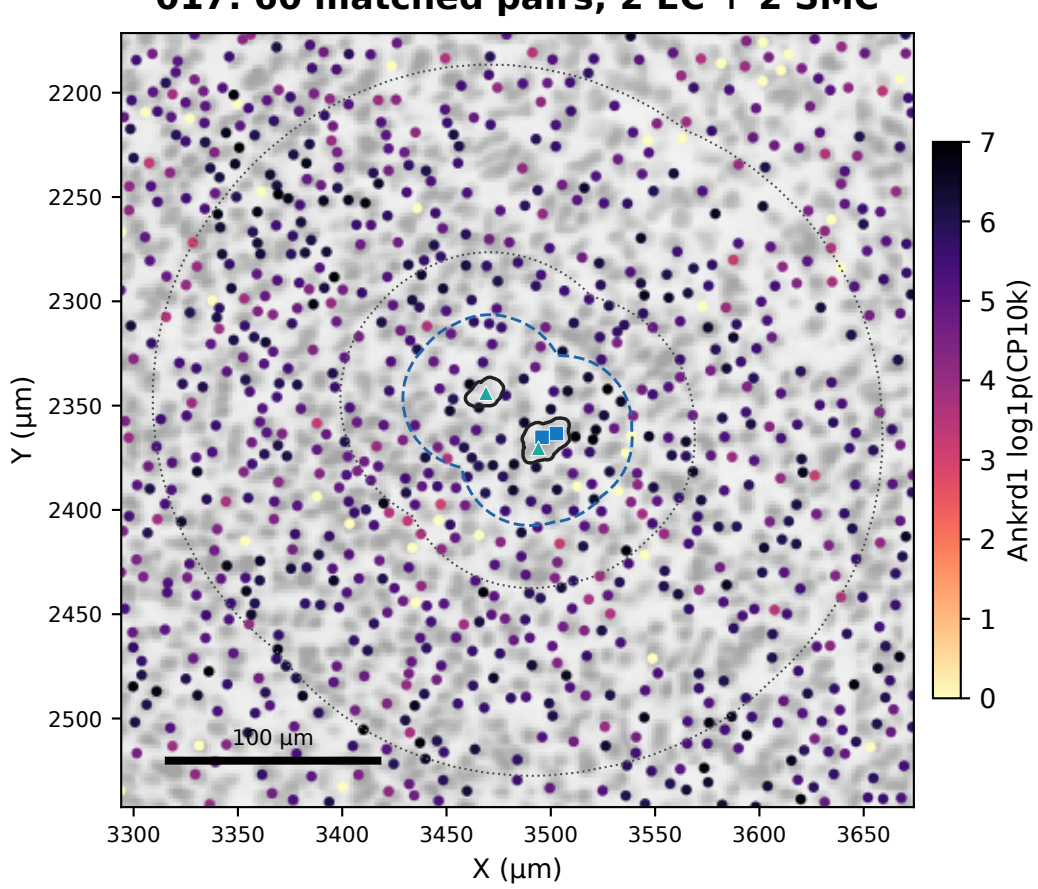
Original cell centroids and DAPI coordinates. All original cells retained; uncertain identities remain flagged.
Arterial candidates require arterial EC + surrounding smooth muscle; outlines are vascular-cell envelopes, not traced vessel walls.
Ankrd1, Creb5, Tlr4 and Atf3 were excluded from all new identity fits and fitting size-factor denominators.

P7: candidate vessel geometry and nearby ventricular CMs





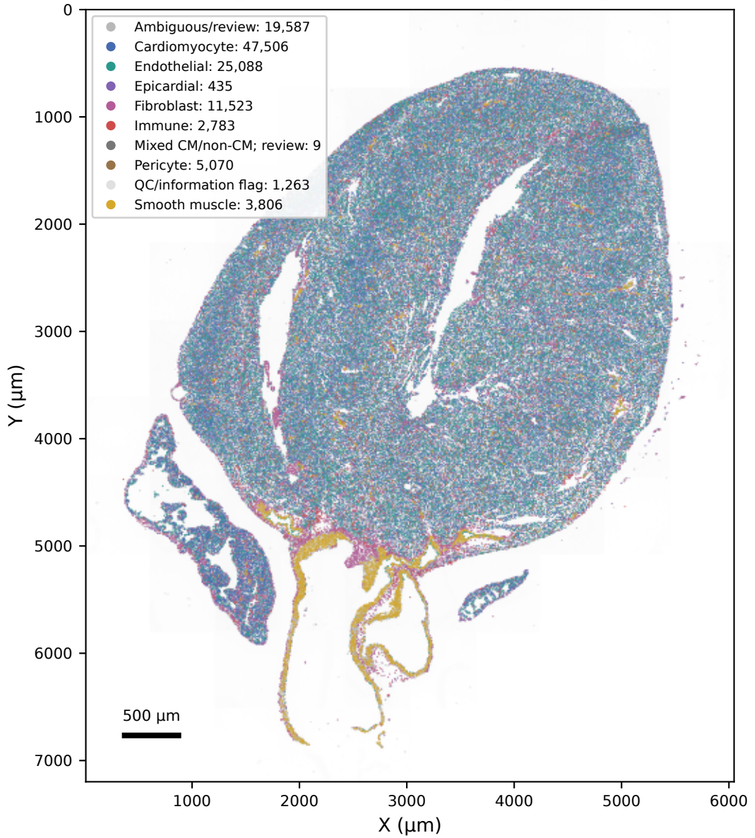




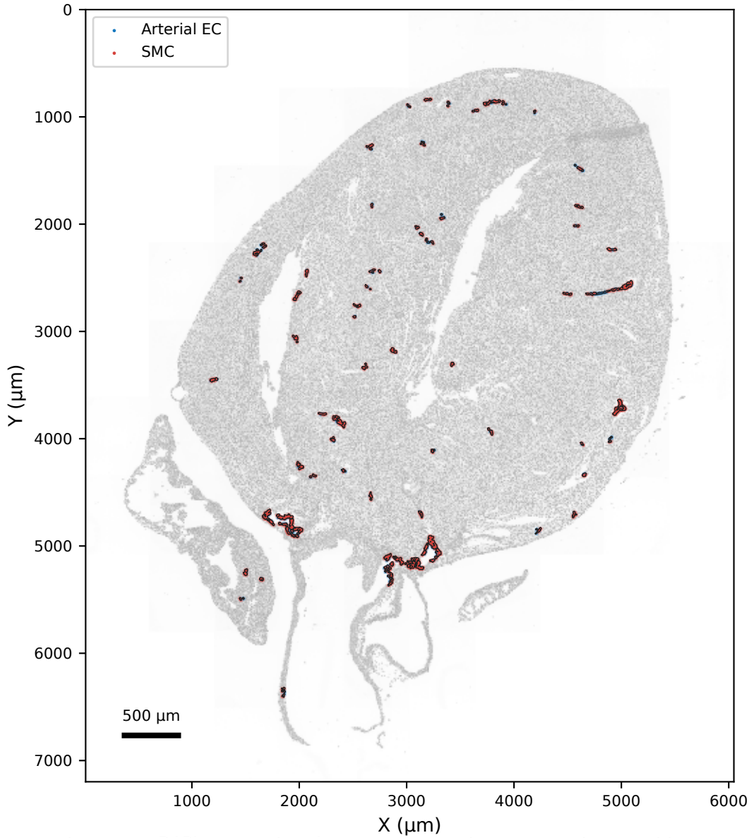
Examples selected by most matched pairs, without selecting on Ankrd1 effects. Square: arterial EC; triangle: SMC.
Black: vascular-cell envelope; blue dashed: 30 µm; gray dotted: 60 and 150 µm.
CM color shows directly measured expression, without smoothing or imputation. Geometry requires anatomical review.

P14: whole-heart reassessment and vascular context

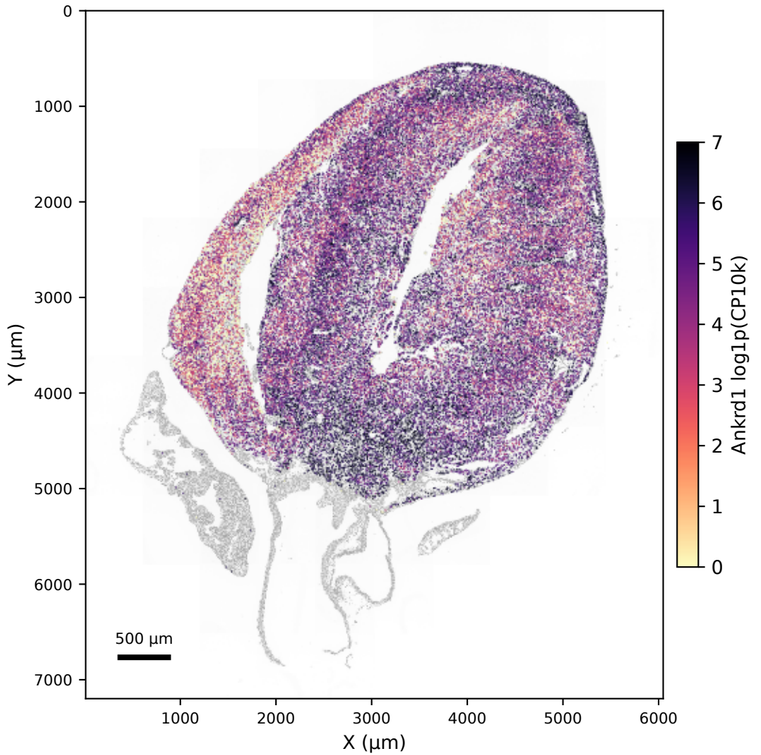
Competing whole-heart lineage assignments



73 candidate small arterial segments

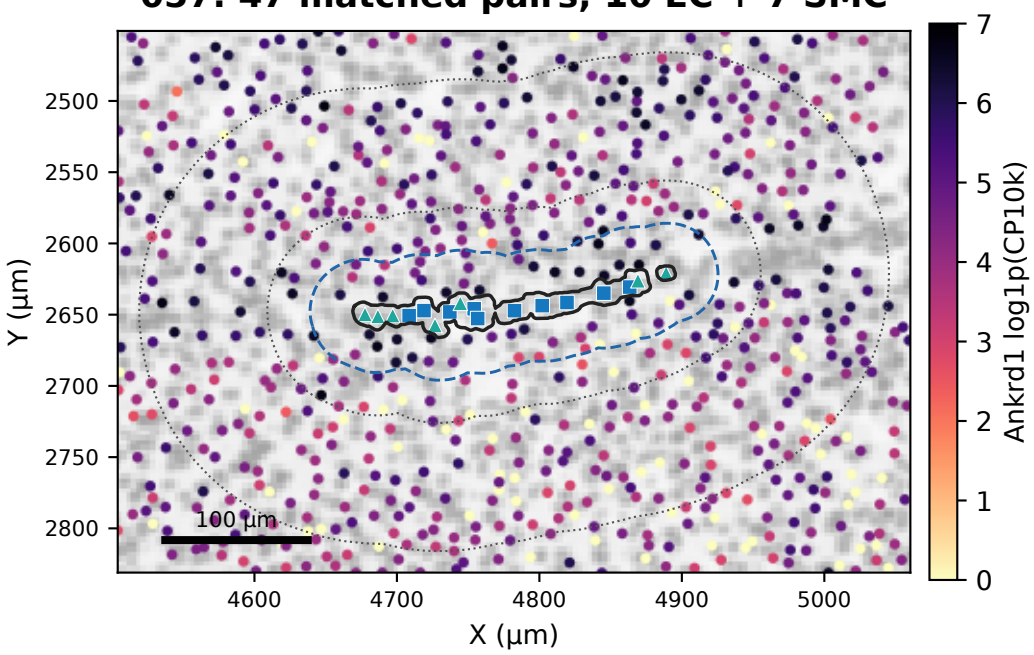


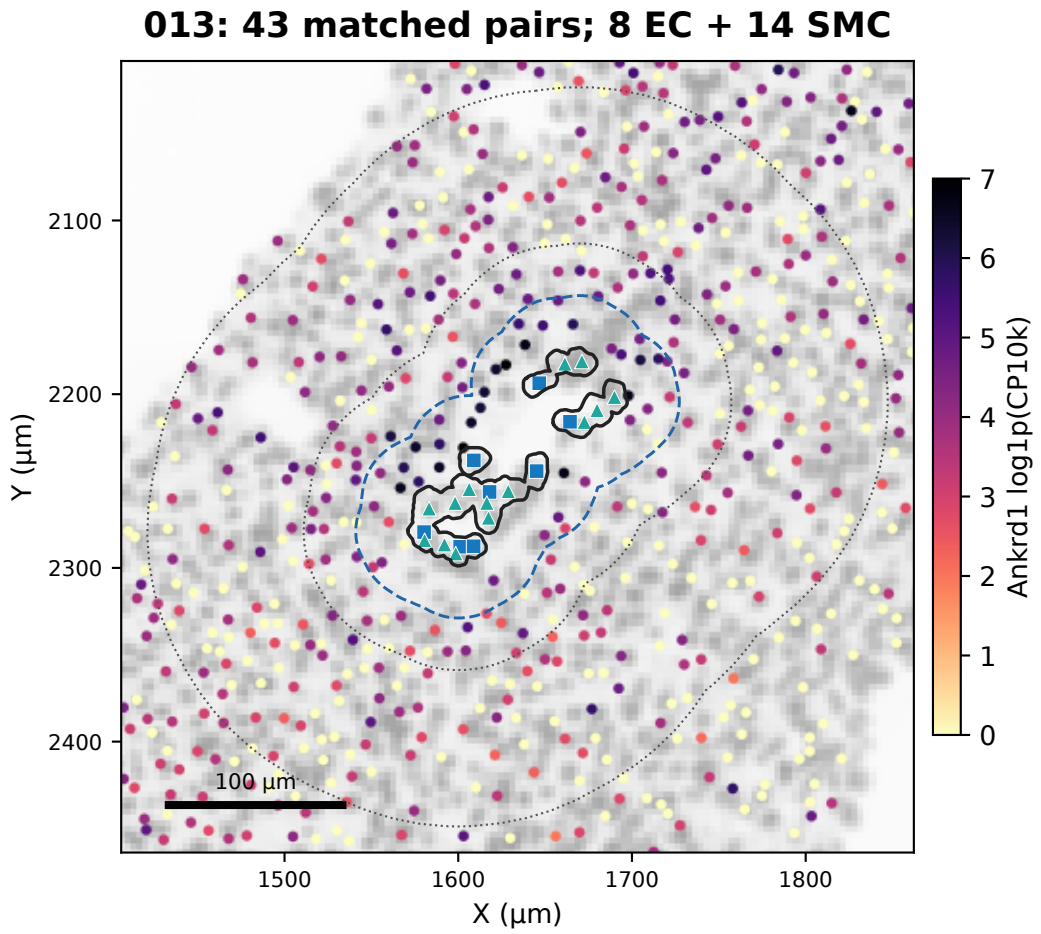
Measured Ankrd1 in strict ventricular CMs

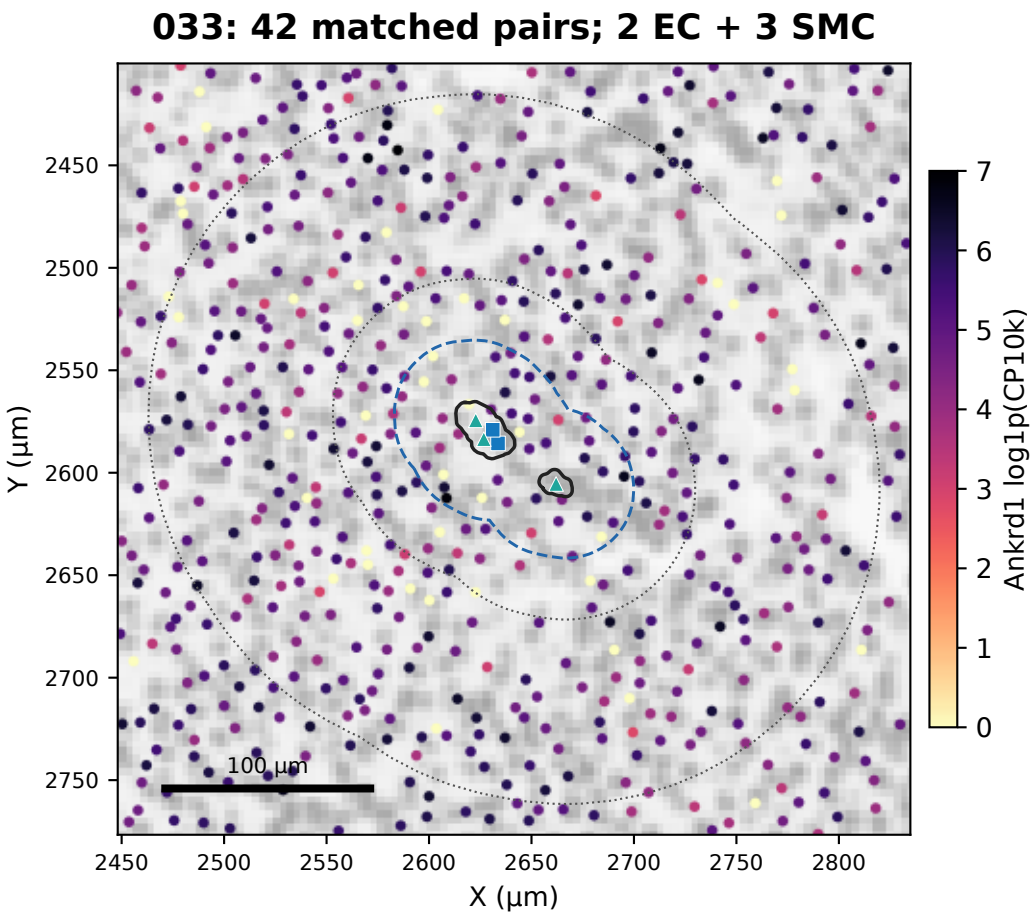


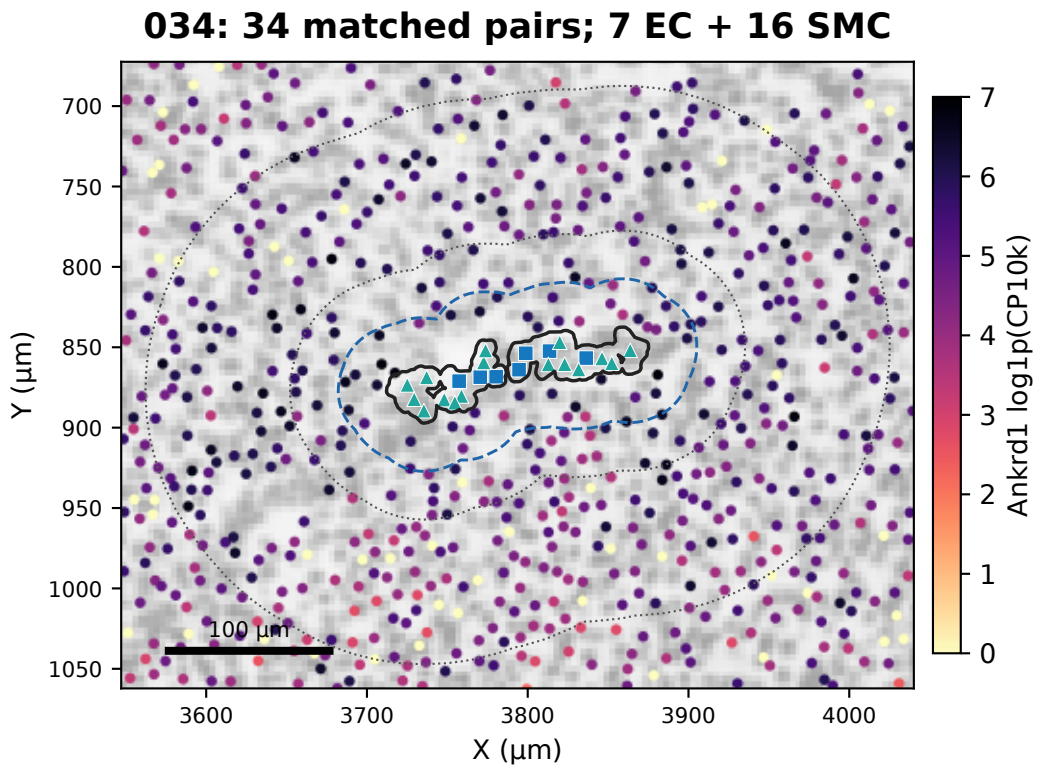
Original cell centroids and DAPI coordinates. All original cells retained; uncertain identities remain flagged.
Arterial candidates require arterial EC + surrounding smooth muscle; outlines are vascular-cell envelopes, not traced vessel walls.
Ankrd1, Creb5, Tlr4 and Atf3 were excluded from all new identity fits and fitting size-factor denominators.

P14: candidate vessel geometry and nearby ventricular CMs



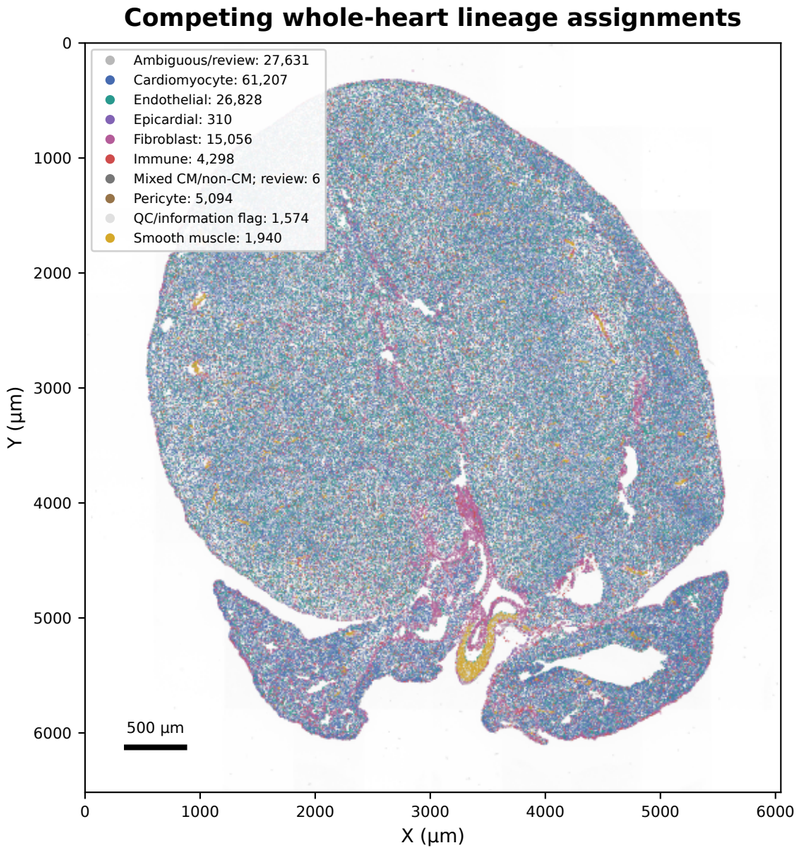


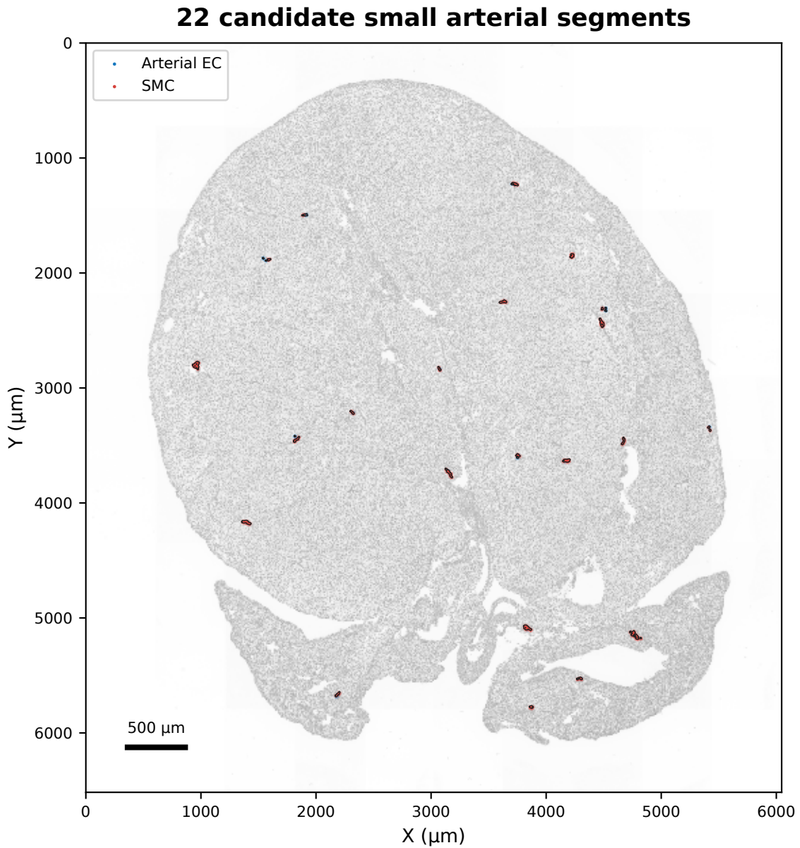


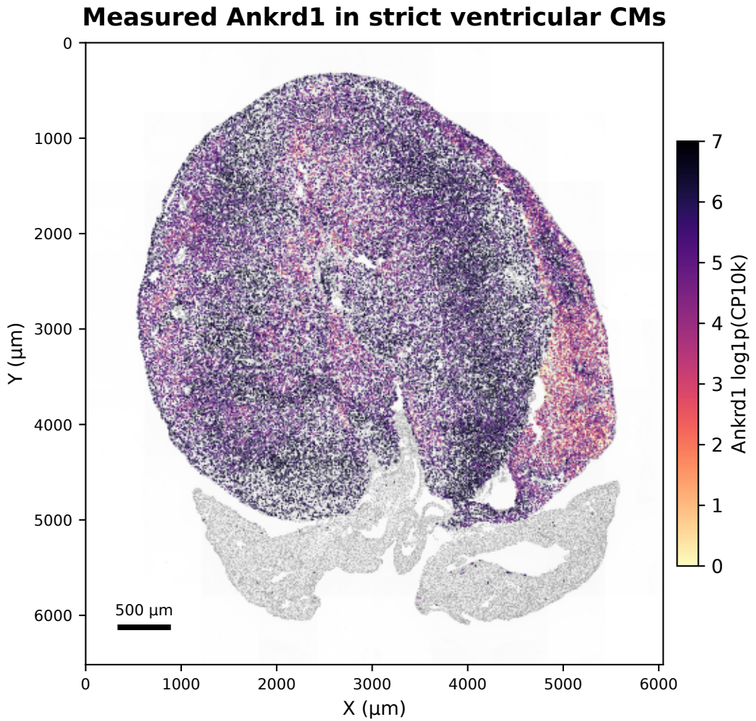


Examples selected by most matched pairs, without selecting on Ankrd1 effects. Square: arterial EC; triangle: SMC.
Black: vascular-cell envelope; blue dashed: 30 µm; gray dotted: 60 and 150 µm.
CM color shows directly measured expression, without smoothing or imputation. Geometry requires anatomical review.

P21: whole-heart reassessment and vascular context

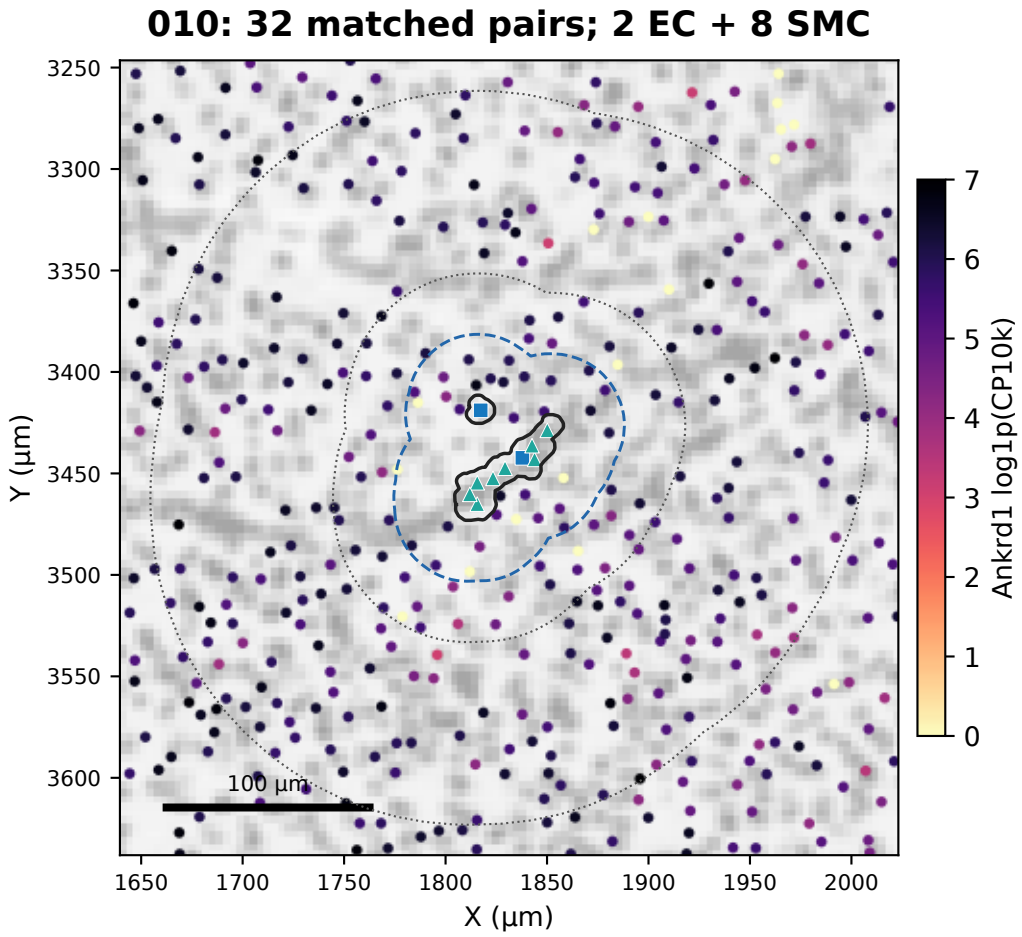


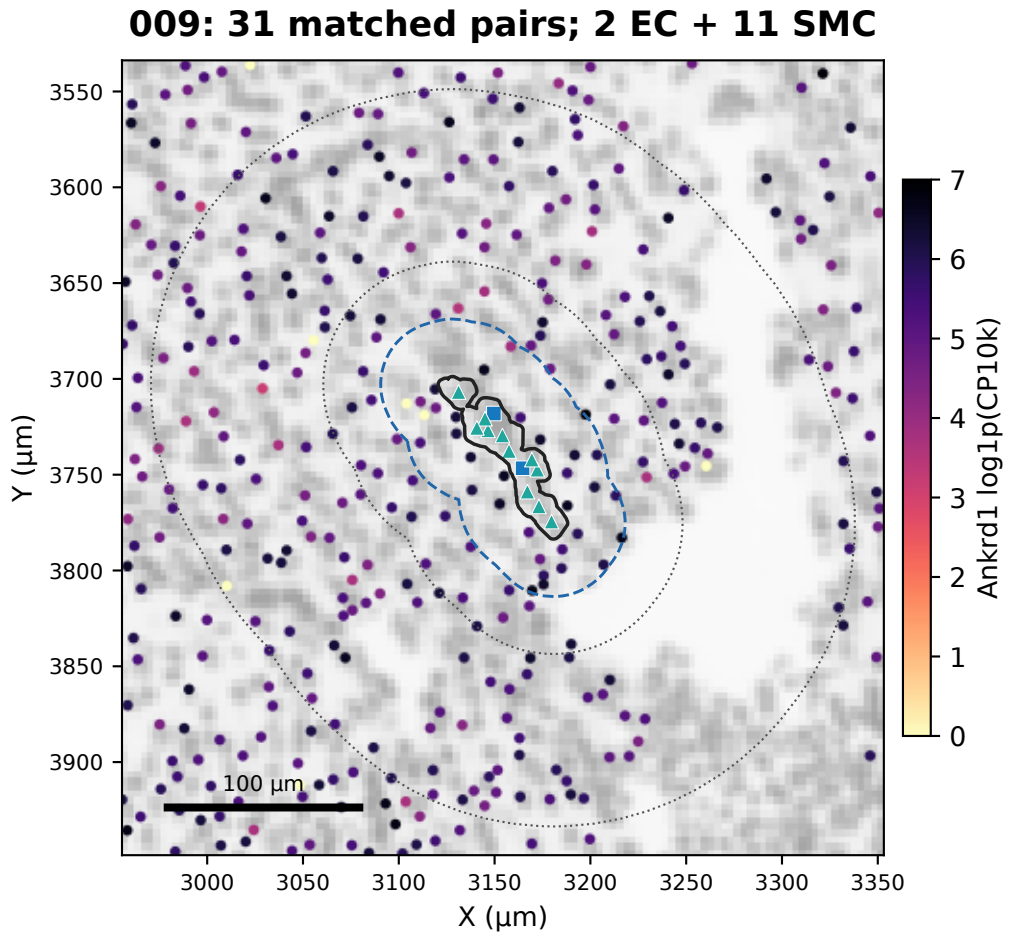


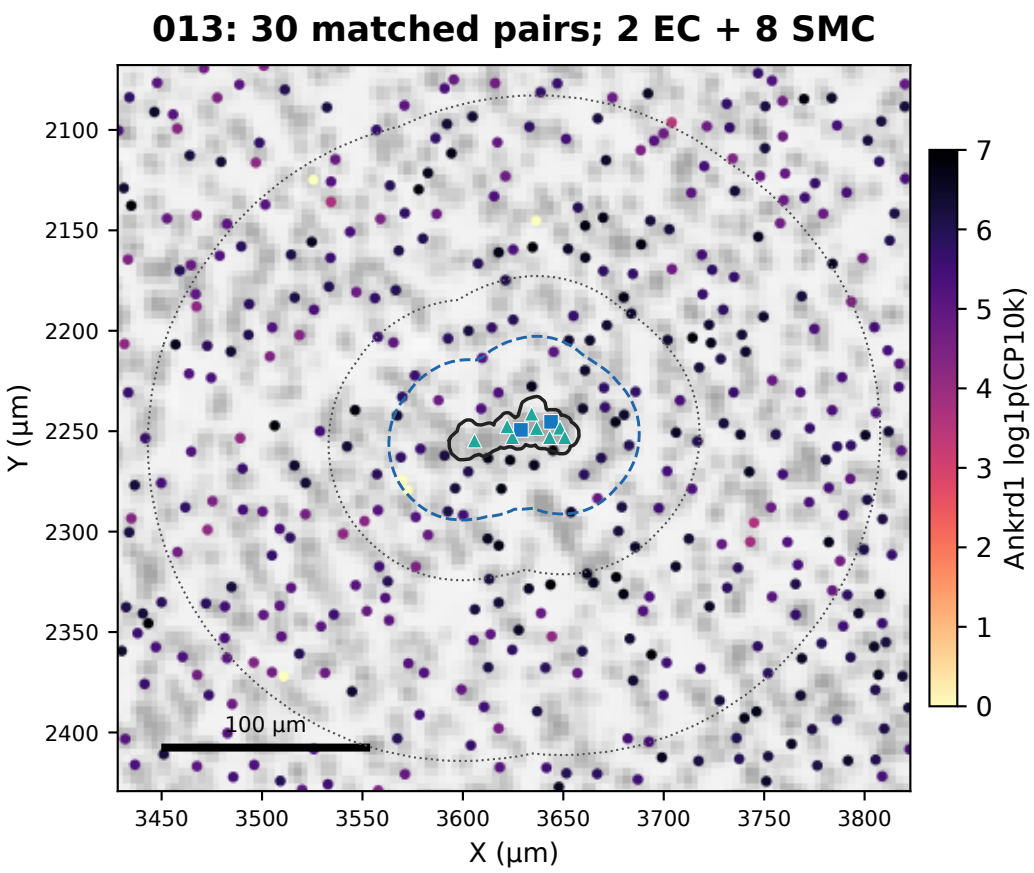


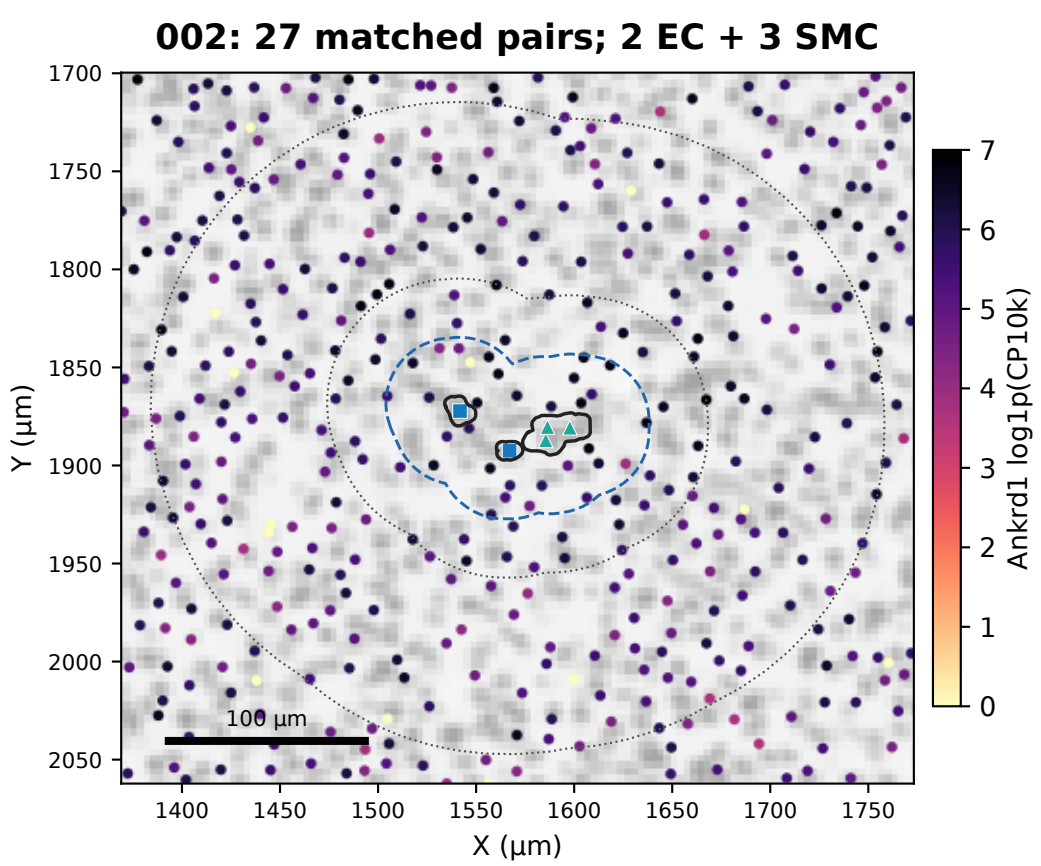
Original cell centroids and DAPI coordinates. All original cells retained; uncertain identities remain flagged.
Arterial candidates require arterial EC + surrounding smooth muscle; outlines are vascular-cell envelopes, not traced vessel walls.
Ankrd1, Creb5, Tlr4 and Atf3 were excluded from all new identity fits and fitting size-factor denominators.

P21: candidate vessel geometry and nearby ventricular CMs



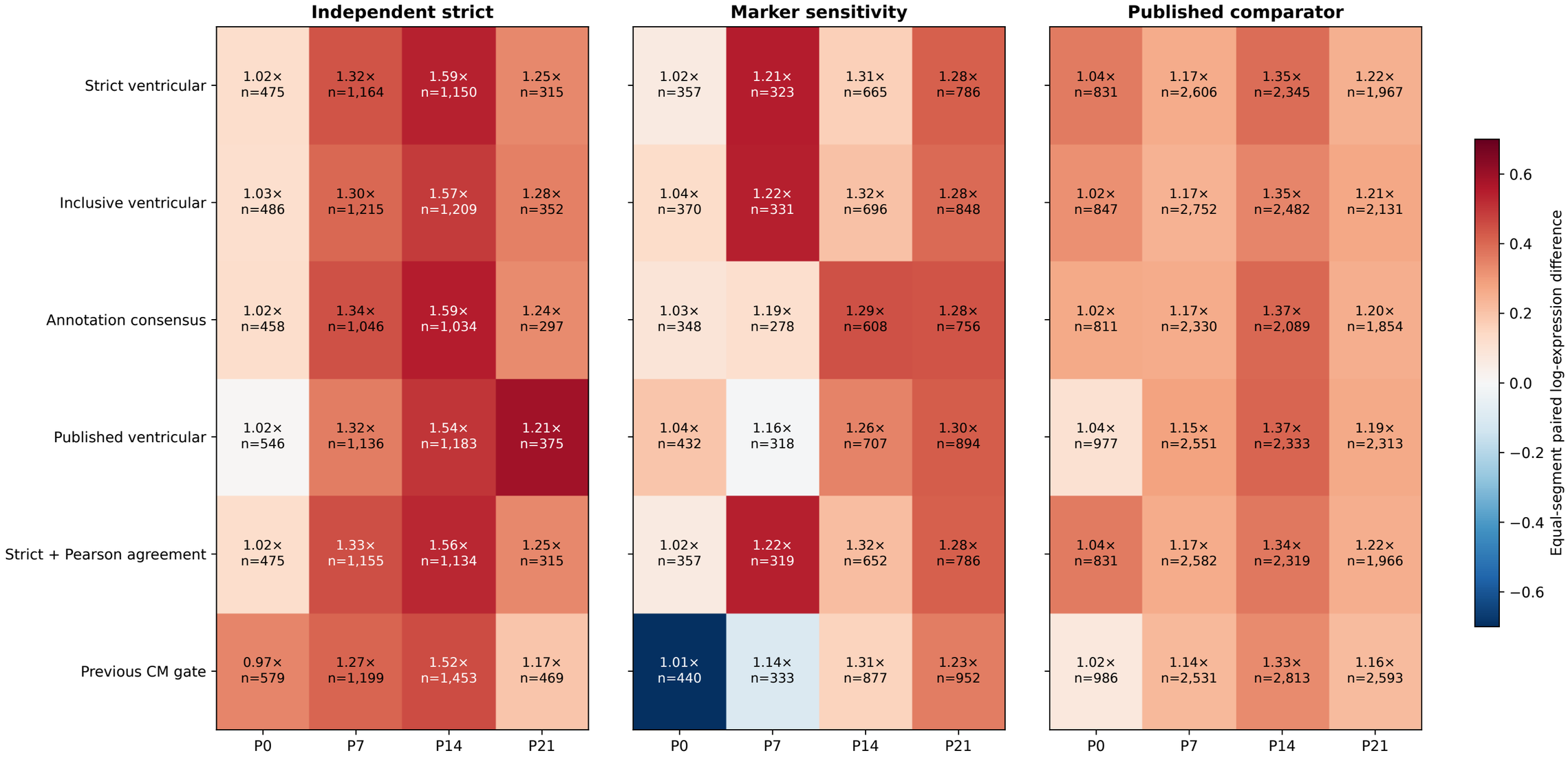






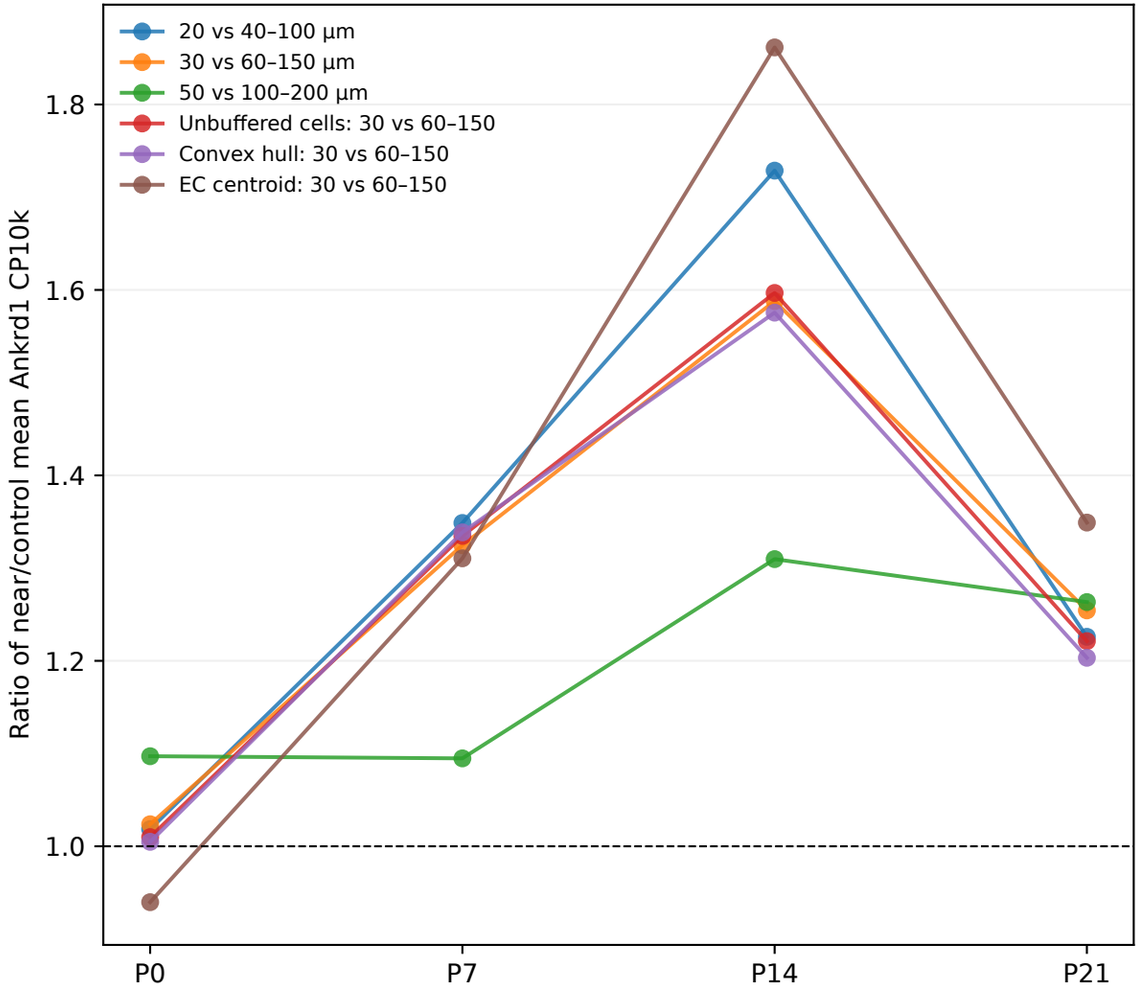
Examples selected by most matched pairs, without selecting on Ankrd1 effects. Square: arterial EC; triangle: SMC.
Black: vascular-cell envelope; blue dashed: 30 µm; gray dotted: 60 and 150 µm.
CM color shows directly measured expression, without smoothing or imputation. Geometry requires anatomical review.

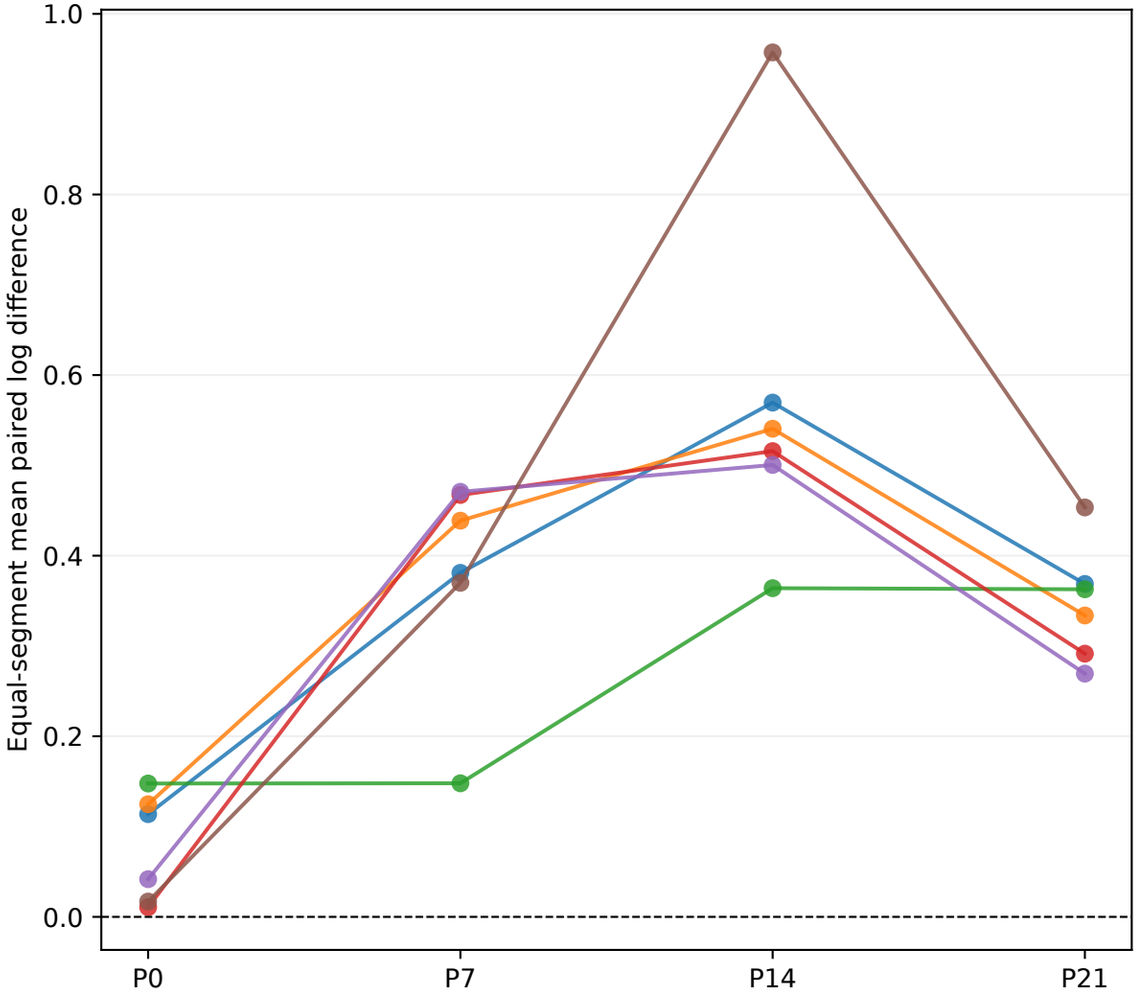
All identity and vessel-definition sensitivities



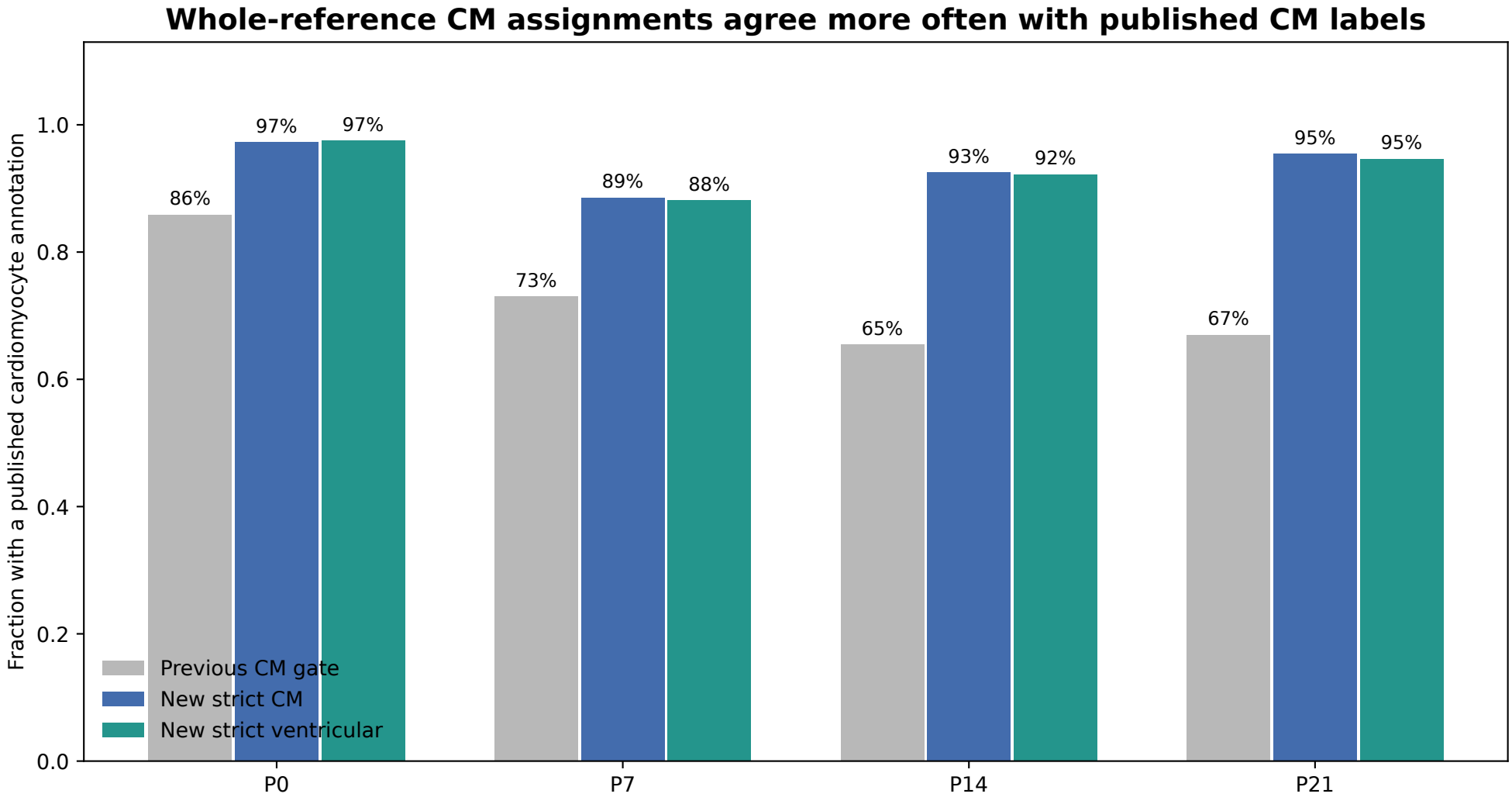
Text: near/control Ankrd1 mean CP10k ratio and matched-pair count. Color: equal-segment mean paired log difference.
Default 30 vs 60–150 µm contrast. Different definitions select different cells and structures.
These are sensitivity analyses of the same four sections, not independent replications.

Distance and geometry sensitivity: strict ventricular CMs





All alternatives shown; vascular membership and small-segment restrictions held fixed.
Distances to envelopes, hulls and EC centroids are different biological approximations; matching cohorts can change.



Agreement is not independently established accuracy; cohort membership and several analysis choices changed.
Published spatial labels were not used to fit the new primary lineage assignments. Ambiguous cells were retained.
The spatial outcome analysis also changes vessel geometry and local controls, so its revision cannot be attributed to lineage mapping alone.